MATH3026/4022 Lecture 13
Time Series using the Frequency Domain I

Dr. Aidan O'Keeffe

School of Mathematical Sciences
University of Nottingham

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Introduction

During previous lectures, we have considered time series in the time domain, typically considering analysis of data of the form $X_1, X_2, \ldots, X_t, X_{t+1}, \ldots$.

Another approach to analysing time series data is to consider a series in the **frequency domain**.

Suppose that a time series exhibits a repeating (e.g. seasonal) pattern.

The **frequency** of a time series is thought of as the number of occurrences of this 'repeating pattern' per unit time.

Analysis of time series using the **frequency domain** may be useful when a time series has obvious periodic components (as many time series do).

Introduction

In applied mathematics, we know that suitable transforms (e.g. a Fourier transform) can be used to transform a function of time, $f(t)$, between the time domain and the frequency domain.

The time series models that we have encountered up to now have used data from the time domain.

Here the term 'data from the time domain' mean that our time series process, written X_t, has been considered (and modelled) as a function of the time t.

We've considered processes of written in the form $\{X_t : t \in \mathcal{T}\}$ where the index set $\mathcal{T}$ represents **time**.

Introduction

Using the **time domain** is not the only approach with which we can consider a time series.

We know that time series data contain variability and, often, exhibit seasonal patterns.

The variability in a process can be represented in terms of periodic components at various frequencies.

First, we'll recap some features of periodic functions.

Periodic Functions

A basic function with one periodic component is

$$f(t) = A\cos(wt + \phi)$$

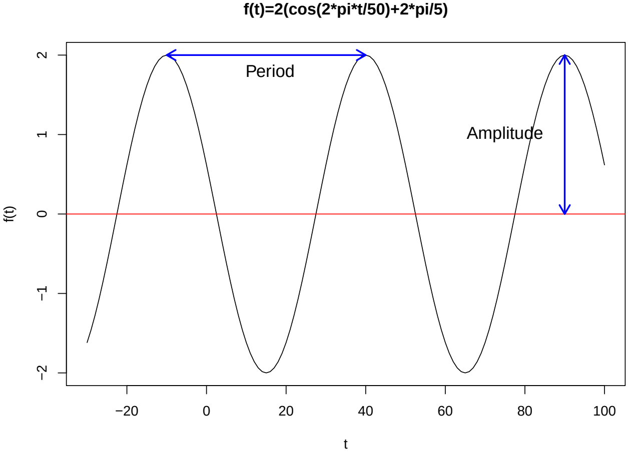
Here

- $A =$ Amplitude, the 'height' of the peak – the maximum deviation from the function's reference value.
- $w =$ Frequency, the number of repetitions of the 'cycle' per unit time.
- $2\pi/w =$ Period, the length of time taken for one 'cycle' of the function.
- $\phi =$ Phase angle, a measure used to 'shift' the function along the time domain (forward or back in time). This does not affect the period, frequency or amplitude.

Periodic Functions

For example,

$$f(t) = 2\cos\left(\frac{2\pi}{50}t + \frac{2\pi}{5}\right).$$



Periodic Functions

Here

$$f(t) = 2\cos\left(\frac{2\pi}{50}t + \frac{2\pi}{5}\right).$$

- $A = 2$, the maximum 'height' of the peak.
- $w = 2\pi/50 =$ Frequency, the number of repetitions of the 'cycle' per unit time.
- $2\pi/w = 50 =$ Period, the length of time taken for one 'cycle' of the function.
- $\phi = \frac{2\pi}{5}$, the phase angle, to set where the cosine function begins.

Periodic Functions

On the previous slide, the function $f(t)$ is a deterministic function.

In time series analysis, we model stochastic processes and, as such, we must consider how to use periodic functions to represent a stochastic process $\{X_t\}$.

As an introductory example, consider the process $\{X_t\}$, defined

$$X_t = A\cos(wt + \phi) + Z_t$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2.

Here, we assume that the period of the periodic component $(p = 2\pi/w)$ is known.

Introductory Example

We can re-write X_t as a linear combination of $\sin$ and $\cos$ terms – this will be a useful approach to take.

Introductory Example

Suppose that we set

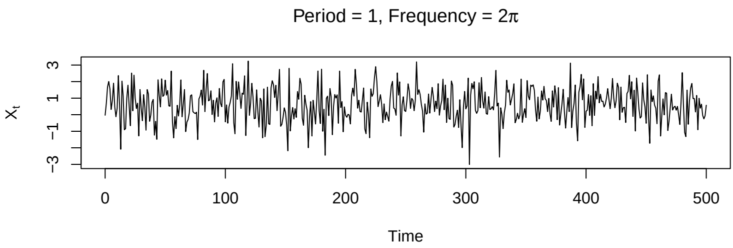
$$\begin{aligned} X_t &= 2\cos\left(\frac{2\pi}{p}t + \frac{2\pi}{5}\right) + Z_t \\ &= 2\cos\left(\frac{2\pi}{p}\left(t + \frac{p}{5}\right)\right) + Z_t \end{aligned}$$

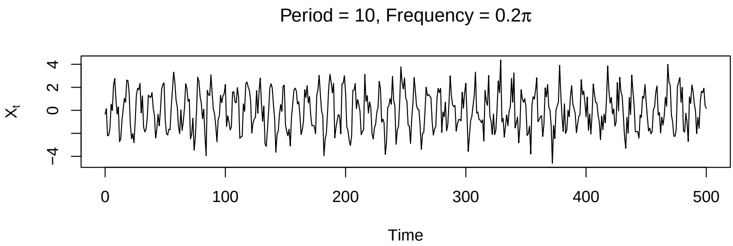
We'll assume that $\{Z_t\}$ is a white noise Gaussian process (i.e. normally distributed) with variance σ_z^2.

Here, the frequency is $2\pi/p$, where p is the period of the function.

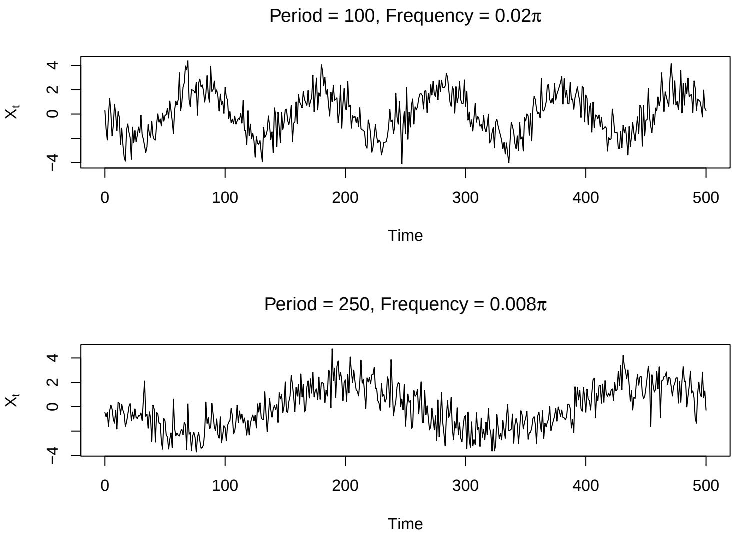
On the next few slides, we consider plots of simulated data from this time series (setting $\sigma_z^2 = 1$), for different periods (short to long).

Introductory Example

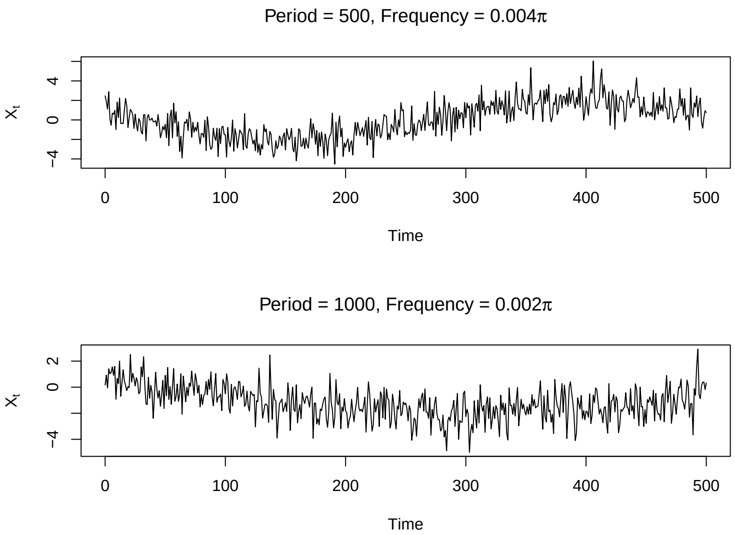




Introductory Example



Introductory Example



Introductory Example

Examining these plots, at each time point we could consider a time series (process) as being constructed as the sum of contributions from various frequencies.

To characterise the process, our aim would be to assess which frequencies are most important in determining the variability in a process.

Some frequencies may contribute to the variability in the time series more than others (and these would influence how the time series changes over time).

Introductory Example

For example, if a high frequency has strong influence on the behaviour of the process then we might expect the series to fluctuate a lot than if low frequencies have more influence on the process.

In fact, a time series can be expressed as a combination of sine and cosine waves with differing periods.

Determining how different sine/cosine waves (and hence which frequencies) contribute to the variability of the series is **analysis in the frequency domain**.

Periodic Functions in Time Series

We can incorporate periodic (sine/cosine) functions into time series analysis.

Suppose that X_t is a variable from a (weakly) stationary time series in the time domain.

We can write X_t as a sum of periodic components, using the form

$$X_t = \sum_{j=0}^{\infty} (P_j \cos w_j t + Q_j \sin w_j t)$$

Here, the P_j and Q_j terms are random variables with properties that we will define and use.

At this point, we have the same type of random process as in previous lectures (written $\{X_t\}$) but our aim will be to determine the relationship between the frequencies $w_1, w_2, w_3, \ldots$ and X_t.

We have

$$X_t = \sum_{j=0}^{\infty} (P_j \cos w_j t + Q_j \sin w_j t)$$

where

$$\begin{aligned} \mathbb{E}(P_j) = \mathbb{E}(Q_j) &= 0; \\ \mathbb{E}(P_j Q_k) &= 0 \ \text{ for all pairs } (j,k); \\ \mathbb{E}(P_j P_k) &= 0 \ \text{ if } j \neq k; \\ \mathbb{E}(Q_j Q_k) &= 0 \ \text{ if } j \neq k. \end{aligned}$$

Periodic Functions in Time Series

In addition

$$Q_0 = 0;$$
$$\mathbb{E}(P_0^2) = g_0;$$
$$\mathbb{E}(P_j^2) = \mathbb{E}(Q_j^2) = 2g_j \quad \text{(where } j \neq 0\text{)}.$$

Now, for $j \neq 0$, we define

$$R_j = \frac{1}{2}\left(P_j - iQ_j\right) \quad \text{(here } i = \sqrt{-1}\text{)}.$$

Then we can write X_t in terms of R_j.

Periodic Functions in Time Series

We see that, for $j \neq 0$

$$P_j \cos w_j t + Q_j \sin w_j t = R_j e^{iw_j t} + \bar{R}_j e^{-iw_j t}.$$

Hence

$$\begin{aligned} X_t &= P_0 + \sum_{j=1}^{\infty} P_j \cos w_j t + Q_j \sin w_j t; \\ &= P_0 + \sum_{j=1}^{\infty} \left(R_j e^{iw_j t} + \bar{R}_j e^{-iw_j t} \right). \end{aligned}$$

In addition, if we define $R_0 = P_0$ then

$$X_t = R_0 + \sum_{j=1}^{\infty} \left(R_j e^{iw_j t} + \bar{R}_j e^{-iw_j t} \right)$$

Now, we define

$$w_0 = 0;$$
$$w_{-j} = -w_j;$$
$$R_{-j} = \bar{R}_j.$$

Then

$$X_t = \sum_{j=-\infty}^{\infty} R_j e^{iw_j t}.$$

We consider the random variables R_j (for $j \neq 0$), recalling that

$$R_j = \frac{1}{2}\left(P_j - iQ_j\right).$$

We observe

$$\mathbb{E}(R_j) = 0;$$
$$\mathbb{E}(|R_j|^2) = g_j.$$

What do these results tell us about X_t?

We know that

$$X_t = \sum_{j=-\infty}^{\infty} R_j e^{iw_j t}.$$

and, since X_t is a linear combination of R_j terms, each with mean zero, we see that $\mathbb{E}(X_t) = 0$.

Other properties of R_j terms can be used to determine features such as the autocovariance function.

Analysis in the Frequency Domain

$$X_t = \sum_{j=-\infty}^{\infty} R_j e^{iw_j t}.$$

At this point, note that X_t (the value of the series/process) at time t is written as a sum of terms that depend on different frequencies $\{w_j\}$.

Some of these frequencies will contribute more to the value (and variability) of X_t than others.

Determining important frequencies that contribute more to the variability will allow us to gain information (learn) about the behaviour of the process $\{X_t\}$.

Autocovariance Function

Let us suppose that our time series consists of information from only a finite number of frequencies, with representation

$$X_t = \sum_{j=-n}^{n} R_j e^{iw_j t}.$$

We consider the autocovariance function of $\{X_t\}$.

Autocovariance Function

Autocovariance Function

Some Notes

We obtain the result

$$\gamma(h) = \sum_{j=-n}^{n} g_j e^{-iw_j h}.$$

We note the following

1. $g_{-j} = \mathbb{E}\left[|R_{-j}|^2\right] = \mathbb{E}\left[|R_j|^2\right] = g_j$ because $R_{-j} = \bar{R}_j$.

   It follows that $\gamma(h) = \gamma(-h)$.

2. $$\begin{aligned}\gamma(h) &= \sum_{j=-n}^{n} g_j e^{-iw_j h} \\ &= g_0 + \sum_{j=1}^{n} g_j (e^{iw_j h} + e^{-iw_j h})\end{aligned}$$

Some Notes

3.

$$\gamma(0) = \sum_{j=-n}^{n} g_j = \mathsf{Var}(X_t)$$
$$= g_0 + 2\sum_{j=1}^{n} g_j.$$

Here, g_j measures the amount of variation in the series due to w_j.

Also, since the values g_j are such that,

$$\gamma(h) = \sum_{j=-n}^{n} g_j e^{-iw_j h}$$

We see that g_j is the coefficient that determines the contribution of the frequency w_j to the autocovariance function at lag h, $\gamma(h)$.

Some Notes

Recall that, for analysis in the frequency domain, we want to determine the contribution of the frequency w_j to the variability in the time series $\{X_t\}$.

We see that determining the coefficients g_j will be important in allowing us to do this.

The values g_j are known as the **spectrum** of a time series.

Usually, for the frequencies, we restrict our attention to

$$-\pi < w_j < \pi.$$

If n is finite then way say that the spectrum is discrete.

Spectral Density

We define the following differences for functions S and G of w_j

$$R_j = S(w_{j+1}) - S(w_j)$$
$$g_j = G(w_{j+1}) - G(w_j)$$

Then

$$X_t = \sum_{j=-n}^{n} R_j e^{iw_j t} = \sum_{j=-n}^{n} \left[S(w_{j+1}) - S(w_j)\right] e^{iw_j t};$$
$$\gamma(h) = \sum_{j=-n}^{n} \left[G(w_{j+1}) - G(w_j)\right] e^{-iw_j h};$$
$$g_j = \mathbb{E}(|R_j|^2).$$

Spectral Density

Suppose that the frequencies $\{w_j\}$ are equally spaced on $(-\pi, \pi]$. Then

$$\begin{aligned} w_j &= j\delta w \quad \text{where } \delta w = 2\pi/n. \\ &= \frac{j2\pi}{n}. \end{aligned}$$

Then, as $n \to \infty$

$$S(w + \delta w) - S(w) = \mathrm{d}S(w)$$
$$G(w + \delta w) - G(w) = \mathrm{d}G(w)$$

and

$$X_t = \int_{-\pi}^{\pi} e^{iwt} \mathrm{d}S(w).$$

Spectral Density

Moreover

$$\begin{aligned}\gamma(h) &= \int_{-\pi}^{\pi} e^{-iwh} \mathrm{d}G(w) \\ &= \int_{-\pi}^{\pi} e^{-iwh} g(w) \mathrm{d}w \quad \text{where } g(w) = \frac{\mathrm{d}G(w)}{\mathrm{d}w}.\end{aligned}$$

Informally

$$g_j \approx g(w)\mathrm{d}w = \mathrm{d}G(w) = \mathbb{E}\left(|\mathrm{d}S(w)|^2\right).$$

Spectral Density Function

$g(w)$ is known as the **spectral density function** of a time series.

$G(w)$ is known as the **spectral distribution function** of a time series.

We note the similarity with probability density/cumulative distribution functions, in that $G'(w) = g(w)$.

The function $g(w)$ with the domain $-\pi \leq w \leq \pi$ is symmetric about 0.

As such, we usually plot a spectral density for $0 \leq w \leq \pi$.

Spectral Distribution Function

The spectral distribution function (or spectrum), $G(w)$, is the unique function on $[-\pi, \pi]$ such that

1. $G(-\pi) = 0$.
2. G is non-decreasing and right-continuous.
3. The increments of G are symmetric about zero. This implies that for $0 \leq a < b \leq \pi$

$$G(b) - G(a) = G(-a) - G(-b).$$

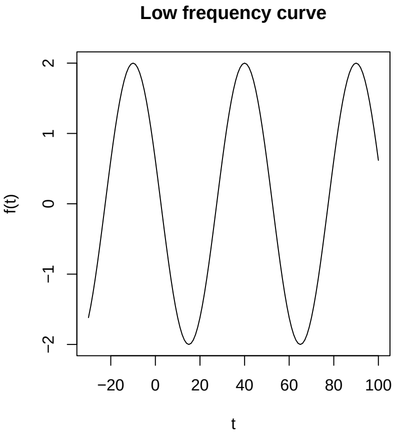
Spectral Density Function

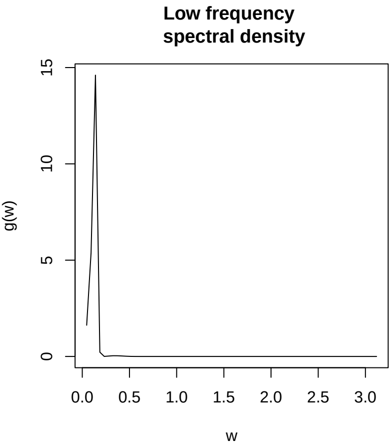
The spectral density function measures the contribution to the total variability of the process at a given frequency.

Mathematically, the autocovariance function and spectral distribution function contain equivalent information about the the process $\{X_t\}$.

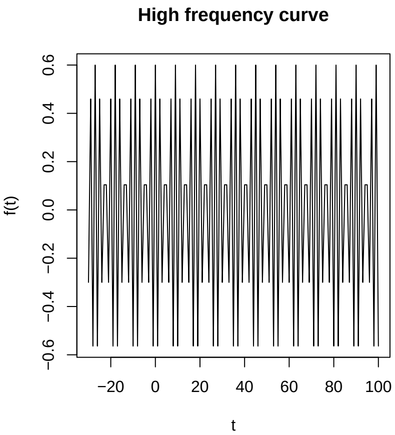
The frequency domain representation (i.e. spectral representation) of a time series may be more interpretable, especially where a process may exhibit periodic variations about the mean.

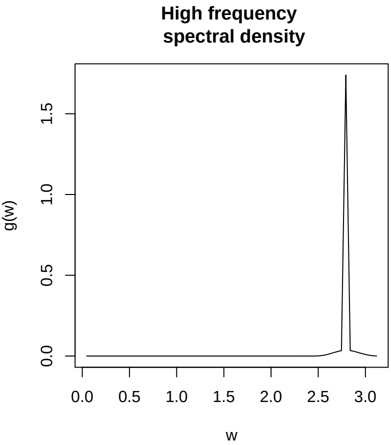
Spectral Density: Example





Spectral Density: Example





Spectral Density Function

In essence, the spectral density function tells us which frequencies have most importance in contributing to the variability of the process $\{X_t\}$

By determining the spectral density function of a time series we will be able to explore properties of the series.

For example, if the spectral density is small around high frequencies but large around low frequencies the series may have short-term positive correlation and then remain smoother in the long term (e.g. an MA(1) process).

Conversely, if the spectral density is large around higher frequencies and small around lower frequencies, the process is less smooth and will tend to fluctuate rapidly around its mean value.

Further properties and examples will be explored during Lecture 14.

Lecture 13: Learning Outcomes

- Generally, understand what is meant by a periodic function and terms such as the: **amplitude**, **period** and **frequency**.
- Understand that a time series can be considered as a random function of sine/cosine terms at different frequencies.
- Understand what is meant by and how to obtain the **spectral density function** and **spectral distribution function** of a time series.
- Understand what the spectral density function can tell us about the behaviour of a process.